3.4 Informal Theoretical Analysis

BayMOTH is built on a gated two-branch architecture. When the inferred task-relatedness is low, it reverts to the 2-OPT acquisition; when the inferred task-relatedness is sufficiently high (γ dependent), it uses a meta-informed lookahead branch that modifies the horizon planning term through the selected virtual environment and the greedy improvement component. Accordingly, rather than comparing BayMOTH to a universally optimal policy, we analyze it relative to an idealized oracle version of itself. This oracle comparator uses the correct branch, correct source-task (when applicable), together with the exact lookahead term, and in particular reduces to 2-OPT on rounds where fallback is the appropriate choice.

To formalize this, let

$$f^\star := \max_{x \in D} f(x), \qquad r_T^{\mathrm{BM}} := f^\star - \max_{1 \le t \le T} f(x_t^{\mathrm{BM}})$$

denote the simple regret of BayMOTH after T evaluations.

At round t, let $\mathcal{X}_t$ denote the candidate set considered by the optimization routine. We write the fallback 2-OPT acquisition as

$$A_t^{\mathrm{2OPT}}(x) := \Lambda_{0,t}(x) + \mathbb{E}_{GP_t}\left[\max_{x' \in \mathcal{X}_t} \Lambda_{1,t}(x'; x)\right],$$

and let $A_t^{\mathrm{meta},\dagger}(x)$ denote the *oracle* meta-informed acquisition, defined as the exact meta-lookahead branch evaluated using the correct source task and exact expectation. Let $g_t^\dagger \in \{0, 1\}$ denote the oracle branch selection decision, where $g_t^\dagger = 0$ selects the fallback (2-OPT) and $g_t^\dagger = 1$ selects the oracle meta branch. The resulting oracle-gated comparator is

$$A_t^\dagger(x) = (1 - g_t^\dagger) A_t^{\mathrm{2OPT}}(x) + g_t^\dagger A_t^{\mathrm{meta},\dagger}(x).$$

Let

$$x_t^\dagger \in \arg\max_{x \in \mathcal{X}_t} A_t^\dagger(x), \qquad r_T^\dagger := f^\star - \max_{1 \le t \le T} f(x_t^\dagger)$$

denote the corresponding oracle-gated simple regret.

BayMOTH itself uses a NCC based branch selection decision $g_t \in \{0, 1\}$. When $g_t = 0$, it optimizes the same fallback 2-OPT branch. When $g_t = 1$, it optimizes a Monte Carlo approximation of the meta branch based on the selected virtual environment. We denote the acquisition actually optimized by BayMOTH as $\hat{A}_t^{\mathrm{BM}}(x)$.

We make the following assumptions.

Assumption 1 (bounded objective). There exists $B > 0$ such that

$$0 \le f(x) \le B, \qquad \forall x \in D.$$

Assumption 2 (meta-vs-fallback branch disagreement probability). Let

$$\mathcal{E}_t^{\mathrm{route}} := \{g_t \neq g_t^\dagger\}$$

denote the event that BayMOTH disagrees with the oracle branch selection rule at round t. Assume

$$\Pr(\mathcal{E}_t^{\mathrm{route}}) \le p_t.$$

Assumption 3 (within-meta branch virtual-environment selection error). For each round t, let $\chi_t^\dagger$ denote the oracle virtual-environment index, i.e., the source task that the oracle meta branch would use when meta-routing is appropriate. Let $\hat{\chi}_t$ denote the virtual-environment index selected by BayMOTH's NCC rule on rounds where $g_t = 1$.

Let

$$A_t^{\mathrm{meta}}(x; \chi)$$

denote the *common exact meta-acquisition template* evaluated using virtual environment GP_χ and with the expectation computed exactly (that is, without Monte Carlo approximation). Then, on rounds where both BayMOTH and the oracle use the meta branch,

$$A_t^{\text{meta},\dagger}(x) = A_t^{\text{meta}}(x;\chi_t^\dagger), \qquad A_t^{\text{meta},\star}(x) = A_t^{\text{meta}}(x;\hat{\chi}_t),$$

and we assume

$$\sup_{x\in\mathcal{X}_t}\left|A_t^{\text{meta}}(x;\hat{\chi}_t) - A_t^{\text{meta}}(x;\chi_t^\dagger)\right| \le \eta_t.$$

If $(g_t, g_t^\dagger) \neq (1,1)$, this assumption is not invoked: the case $g_t \neq g_t^\dagger$ is handled by Assumption 2, and when $g_t = g_t^\dagger = 0$, both methods use the same fallback 2-OPT branch.

Assumption 4 (Monte Carlo approximation error). Let $\hat{J}_{t,M}(x)$ denote the Monte Carlo estimator used by BayMOTH in the meta branch with M samples. Assume

$$\mathrm{Var}(\hat{J}_{t,M}(x)) = \frac{\sigma_t^2(x)}{M},$$

and define

$$\sigma_t := \sup_{x\in\mathcal{X}_t}\sigma_t(x).$$

Assumption 5 (acquisition-to-value stability). There exists $L_t > 0$ such that for any two acquisitions A_t and $\tilde{A}_t$ on $\mathcal{X}_t$,

$$\sup_{x\in\mathcal{X}_t}|A_t(x) - \tilde{A}_t(x)| \le \varepsilon \quad\Longrightarrow\quad f(x_{\tilde{A}_t}) - f(x_{A_t}) \le L_t\varepsilon,$$

where $x_{A_t} \in \arg\max_{x\in\mathcal{X}_t} A_t(x)$ and similarly for $x_{\tilde{A}_t}$.

Proposition 1 (Informal simple-regret decomposition for BayMOTH)**.** *For any sequence $\{\delta_t\}_{t=1}^T$ with $\delta_t \in (0,1)$, with probability at least*

$$1 - \sum_{t=1}^T |\mathcal{X}_t|\delta_t - \sum_{t=1}^T p_t,$$

the simple regret of BayMOTH satisfies

$$r_T^{\text{BM}} \le r_T^\dagger + \max_{1\le t\le T} g_t^\dagger L_t\left(\eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}\right).$$

Moreover,

$$\mathbb{E}[r_T^{\text{BM}}] \le \mathbb{E}[r_T^\dagger] + \max_{1\le t\le T} g_t^\dagger L_t\left(\eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}\right) + B\left(\sum_{t=1}^T|\mathcal{X}_t|\delta_t + \sum_{t=1}^T p_t\right).$$

Proof sketch. Let $\mathcal{G}_T$ denote the event that, for every round $t \le T$, BayMOTH agrees with the oracle branch decision We first work on the event $\mathcal{G}_T$.

Fix a round t. Since $g_t = g_t^\dagger$ on $\mathcal{G}_T$, there are two cases.

If $g_t^\dagger = 0$, both BayMOTH and the oracle comparator optimize the same fallback acquisition A_t^{2OPT}, and hence there is no acquisition discrepancy at round t.

If $g_t^\dagger = 1$, both procedures use the meta branch, and the discrepancy can be decomposed as

$$\hat{A}_t^{\text{BM}}(x) - A_t^\dagger(x) = \underbrace{\left(\hat{A}_t^{\text{BM}}(x) - A_t^{\text{meta},\star}(x)\right)}_{\text{Monte Carlo error}} + \underbrace{\left(A_t^{\text{meta},\star}(x) - A_t^{\text{meta},\dagger}(x)\right)}_{\text{virtual-environment mismatch}}.$$

Under Assumption 3,

$$\sup_{x\in\mathcal{X}_t}\left|A_t^{\text{meta},\star}(x) - A_t^{\text{meta},\dagger}(x)\right| \le \eta_t.$$

For the Monte Carlo term, Assumption 4 and Chebyshev's inequality imply that for each fixed $x \in \mathcal{X}_t$,

$$\Pr\left(|\hat{J}_{t,M}(x) - J_t^{\text{meta},\star}(x)| \geq \frac{\sigma_t}{\sqrt{M\delta_t}}\right) \leq \delta_t.$$

Equivalently, for each fixed $x \in \mathcal{X}_t$, with probability at least $1 - \delta_t$,

$$|\hat{J}_{t,M}(x) - J_t^{\text{meta},\star}(x)| \leq \frac{\sigma_t}{\sqrt{M\delta_t}}.$$

Now define the bad event

$$\mathcal{B}_t := \left\{\exists x \in \mathcal{X}_t : |\hat{J}_{t,M}(x) - J_t^{\text{meta},\star}(x)| > \frac{\sigma_t}{\sqrt{M\delta_t}}\right\}.$$

By the union bound,

$$\Pr(\mathcal{B}_t) \leq \sum_{x \in \mathcal{X}_t} \Pr\left(|\hat{J}_{t,M}(x) - J_t^{\text{meta},\star}(x)| > \frac{\sigma_t}{\sqrt{M\delta_t}}\right) \leq |\mathcal{X}_t|\,\delta_t.$$

Hence, with probability at least $1 - |\mathcal{X}_t|\delta_t$,

$$\forall x \in \mathcal{X}_t, \qquad |\hat{J}_{t,M}(x) - J_t^{\text{meta},\star}(x)| \leq \frac{\sigma_t}{\sqrt{M\delta_t}}.$$

Since the Monte Carlo approximation enters through the acquisition, this implies

$$\sup_{x \in \mathcal{X}_t} \left|\hat{A}_t^{\text{BM}}(x) - A_t^{\text{meta},\star}(x)\right| \leq \frac{\sigma_t}{\sqrt{M\delta_t}}.$$

Therefore, on $\mathcal{G}_T$,

$$\sup_{x \in \mathcal{X}_t} \left|\hat{A}_t^{\text{BM}}(x) - A_t^{\dagger}(x)\right| \leq \sup_{x \in \mathcal{X}_t} \left|\hat{A}_t^{\text{BM}}(x) - A_t^{\text{meta},\star}(x)\right| + \sup_{x \in \mathcal{X}_t} \left|A_t^{\text{meta},\star}(x) - A_t^{\text{meta},\dagger}(x)\right| \leq \eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}.$$

Multiplying by $g_t^\dagger$ makes the bound valid uniformly over both fallback and meta rounds:

$$\sup_{x \in \mathcal{X}_t} \left|\hat{A}_t^{\text{BM}}(x) - A_t^{\dagger}(x)\right| \leq g_t^\dagger \left(\eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}\right).$$

Assumption 5 then transfers this to an objective-value difference:

$$f(x_t^\dagger) - f(x_t^{\text{BM}}) \leq g_t^\dagger L_t \left(\eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}\right).$$

Since simple regret depends on the best value attained over all rounds, we obtain on $\mathcal{G}_T$

$$r_T^{\text{BM}} \leq r_T^\dagger + \max_{1 \leq t \leq T} g_t^\dagger L_t \left(\eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}\right).$$

Finally, by Assumption 2 and a union bound over rounds,

$$\Pr(\mathcal{G}_T^c) \leq \sum_{t=1}^{T} |\mathcal{X}_t|\delta_t + \sum_{t=1}^{T} p_t,$$

which gives the stated high-probability bound.

For the expectation bound, decompose over $\mathcal{G}_T$ and $\mathcal{G}_T^c$. On $\mathcal{G}_T$, the above regret bound holds. On $\mathcal{G}_T^c$, Assumption 1 gives the crude bound $r_T^{\text{BM}} \leq B$. This yields

$$\mathbb{E}[r_T^{\text{BM}}] \leq \mathbb{E}[r_T^\dagger] + \max_{1 \leq t \leq T} g_t^\dagger L_t \left(\eta_t + \frac{\sigma_t}{\sqrt{M\delta_t}}\right) + B\left(\sum_{t=1}^{T} |\mathcal{X}_t|\delta_t + \sum_{t=1}^{T} p_t\right).$$

□

The proposition should be interpreted as a regret decomposition rather than as a sharp problem-dependent theorem. It shows that BayMOTH tracks an oracle-gated version of its own two-branch architecture. In particular, BayMOTH differs from it only through three BayMOTH-specific sources of error: meta-vs-fallback branch disagreement, mismatch between the selected virtual environment and the oracle source model on meta-routed rounds, and Monte Carlo approximation error in the lookahead term.